

SPIDER

447 series
218 patients
4 hospitals

expert
masks

vendor,
field

R1, R2 intensity

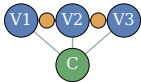
raw signal in masks;
ratios, z scores

R3 mask geometry

volumes, spacing,
counts, voxel size

R4 graph-relational

Laplacian spectrum
+ scale-free ratios



V vertebra
D disc
C canal

Parallel analyses (same protocol)

Scanner predictability

random forest,
grouped CV,
bootstrap,
permutation
on R1 to R4

Acquisition controls

voxel size,
coverage, sequence,
fixed field,
case mix, ComBat
on R3, R4

Downstream utility

degeneration,
narrowing;
within and
across vendors
on R3, R4

Series network

k-NN graph,
assortativity,
communities,
patient null
on R3